

Programming for Biomedical Informatics

Week 3 - Data Integration & Summary Analysis

This week we're first going to practice using a range of the eUtilities end-points to search, fetch, and link data.

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In [2]: # Preliminaries
from Bio import Entrez
import urllib.request
import json
import xml.etree.ElementTree as ET

# ElementTree is a great library for handling and querying XML objects
# https://docs.python.org/3/library/xml.etree.elementtree.html

# load my API key from the file
with open('../api_keys/ncbi.txt', 'r') as file:
    api_key = file.read().strip()

with open('../api_keys/ncbi_email.txt', 'r') as file:
    email = file.read().strip()

Entrez.api_key = api_key
Entrez.email = email
```

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In [ ]: # Step One - Example Using eInfo

# let's use biopython Entrez module to do an eInfo query
# this will tell us what databases are available
handle = Entrez.einfo()

# read the handle
record = Entrez.read(handle)

# print the record in an easy to read format
# print(json.dumps(record, indent=4))

# print out some useful information about each of these databases
# NB this is incredibly useful but long so best to do it for a particular db
# for db in record['DbList']:
#     print(f"Database: {db}")
#     # get the database info
#     db_info = Entrez.read(Entrez.einfo(db=db))
#     # print the database info
#     print(json.dumps(db_info, indent=4))

# let's do this just for the gene database
gene_info = Entrez.read(Entrez.einfo(db='gene'))

# this is a nice way to print out nested XML structures in a readable way
print(json.dumps(gene_info, indent=4))
```

```
In [ ]: # Step Two – Example using eSearch to search the gene database for a particu
# search for the gene 'BRCA1'
handle = Entrez.esearch(db='gene', term='BRCA1', retmode='xml')

#retrieve the record
record = Entrez.read(handle)

# find the count of records
count = record['Count'] # type: ignore
print(f"Number of records found: {count}")

# why are there so many?
```

Number of records found: 34471

```
In [ ]: # print the record in an easy to read format
print(json.dumps(record, indent=4))

# NB the [All Fields] search is a broad search that will return many results
```

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In [6]: # Step Three – Example using eSummary to get some metadata about the first 5
# this time we're going to do this using urllib.requests to show how tou can
# we will request xml and parse that using ElementTree

# get the first 5 ids
ids = record['IdList'][:5] # type: ignore

eUtils_base = "https://eutils.ncbi.nlm.nih.gov/entrez/eutils/"
eSummary = "esummary.fcgi"

# don't forget to add the API key and email
#for each id in the list pull the summary
for id in ids:
    url = f"{eUtils_base}{eSummary}?db=gene&id={id}&api_key={api_key}&email="
    with urllib.request.urlopen(url) as response:
        xml = response.read()
        root = ET.fromstring(xml)
        # find the <Organsim> tag
        organism = root.find('DocumentSummarySet/DocumentSummary/Organism/')
        name = root.find('DocumentSummarySet/DocumentSummary/').text # type:
        print(id,name,organism)

# OK what's going on here?
```

```
143861250 LOC143861250 Tasmanian lanceolata
143856652 LOC143856652 Tasmanian lanceolata
143839474 BABAM2 Paroedura picta
143832220 UIMC1 Paroedura picta
143832066 BAP1 Paroedura picta
```

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In [ ]: # Step Four – Lets look again but add the [Gene] field to the search
handle = Entrez.esearch(db='gene', term='BRCA1[Gene]', retmode='xml')

#retrieve the record
record = Entrez.read(handle)
```

```

# find the count of records
count = record['Count']
print(f"Number of records found: {count}")

# print the record in an easy to read format
print(json.dumps(record, indent=4))

```

```

In [8]: # get the first 5 ids
ids = record['IdList'][:5]

eUtils_base = "https://eutils.ncbi.nlm.nih.gov/entrez/eutils/"
eSummary = "esummary.fcgi"

# don't forget to add the API key and email
# for each id in the list pull the summary
for id in ids:
    url = f"{eUtils_base}{eSummary}?db=gene&id={id}&api_key={api_key}&email={email}"
    with urllib.request.urlopen(url) as response:
        xml = response.read()
        root = ET.fromstring(xml)
        # find the <Organsim> tag
        organism = root.find('DocumentSummarySet/DocumentSummary/Organism/')
        name = root.find('DocumentSummarySet/DocumentSummary/').text
        print(id, name, organism)

# Ah that's better, now we've got BRCA1 but from lots of different organisms

```

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143892821 BRCA1 Tasmanian lanceolata
143825099 BRCA1 Paroedura picta
143765407 BRCA1 Ranitomeya variabilis
143687345 BRCA1 Tamandua tetradactyla
143409333 Brca1 Callospermophilus lateralis

```

```

In [ ]: # Step Five – Finally, let's get this right

# search for the gene 'BRCA1' in the human genome using eSearch
handle = Entrez.esearch(db='gene', term='BRCA1[Gene], human[Organism]', retr

# retrieve the record
record = Entrez.read(handle)

# find the count of records
count = record['Count']
print(f"Number of records found: {count}")

# print the record in an easy to read format
print(json.dumps(record, indent=4))

# get the only id
id = record['IdList'][0]

```

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In [10]: # now use eSummary to get the metadata for this gene
url = f"{eUtils_base}{eSummary}?db=gene&id={id}&api_key={api_key}&email={email}"
response = urllib.request.urlopen(url)
root = ET.fromstring(response.read())
organism = root.find('DocumentSummarySet/DocumentSummary/Organism/').text

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```
name = root.find('DocumentSummarySet/DocumentSummary/').text
print(id,name,organism)
```

672 BRCA1 Homo sapiens

```
In [ ]: # Step Six - Use eFetch to get the full record

# lets modify the code above to use eFetch to get the full record
eFetch = "efetch.fcgi"

url = f"{eUtils_base}{eFetch}?db=gene&id={id}&api_key={api_key}&email={email}"
response = urllib.request.urlopen(url)
xml = response.read()

# print the xml
print(xml.decode('utf-8'))
```

```
In [ ]: # Step Seven - Use eLink to get the associated nucleotide sequence

# lets go back to using the Entrez module to get the sequence
# get the link specifically for the refseq gene nucleotide sequence
links = Entrez.read(Entrez.elink(dbfrom='gene', id=id, linkname='gene_nuclor

# get the id of the nucleotide sequence
nuccore_id = links[0]['LinkSetDb'][0]['Link'][0]['Id']

# get the sequence
handle = Entrez.efetch(db='nuccore', id=nuccore_id, rettype='fasta', retmode=
sequence = handle.read()

# print the sequence
print(sequence)
```